MODELLING OF SOLAR ENERGY HARVESTING USING A PHASE CHANGE MATERIAL ASSISTED SOLAR TRACKING SYSTEM

A PROJECT REPORT

Submitted by

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SANJAI S	**8115U24ME048**
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SRIDHARAN V	**8115U24ME304**

in partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING

IN

MECHANICAL ENGINEERING

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
SAMAYAPURAM, TRICHY

ANNA UNIVERSITY

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MEB1233 DESIGN THINKING

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IN MECHANICAL ENGINEERING

Under the Guidance of

Dr. B. PRAKASH, Ph.D.,

Department of Mechanical Engineering

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

MECHANICAL ENGINEERING

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

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BONAFIDE CERTIFICATE

Certified that this project report titled "**MODELLING OF SOLAR ENERGY HARVESTING USING A PHASE CHANGE MATERIAL ASSISTED SOLAR TRACKING SYSTEM"** is the bonafide work of **MAHAARAJAN C, (8115U24ME033), SANJAI S, (8115U24ME048),YASHWANTH S, (8115U24ME061), SRIDHARAN V , (8115U24ME304),** who carried out the work under my supervision.

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ANNA UNIVERSITY, CHENNAI

DECLARATION BY THE CANDITATE

I declare that to the best of my knowledge the work reported here in has been composed solely by myself and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva- Voce held at K. Ramakrishnan College of Engineering on _____

1.
2.
3.
4.

SIGNATURE OF THE CANDITATE

ACKNOWLEDGMENT

We thank the almighty god without his blessing it would not have been possible for us to complete this project.

At this moment of having successfully completed our project, we wish to convey our sincere thanks and gratitude to our management of our college and our beloved chairman **Dr. K. RAMAKRISHNAN, B.E., Ph.D., who** provide all the facilities to us.

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Dr**. S. KUPPUSAMY, M.B.A., Ph.D**., for forwarding us to do our project and offering adequate duration in completing our project.

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SAMAYAPURAM, TIRUCHIRAPPALLI - 112

VISION OF THE K. RAMAKRISHNAN COLLEGE OF ENGINEERING

- To achieve a prominent position among the top technical institutions

MISSION OF THE K. RAMAKRISHNAN COLLEGE OF ENGINEERING

- To bestow standard technical education par excellence through state of the art infrastructure, competent faculty and high ethical standards.
- To nurture research and entrepreneurial skills among students in cutting edge technologies
- To provide education for developing high quality professionals to transform the society

VISION OF THE MECHANICAL ENGINEERING DEPARTMENT

- To provide eminent Mechanical Engineers to the society and industries through quality technical education and research.

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- To motivate students and faculty to undertake research and development activities in the thrust areas of mechanical engineering.
- To carryout collaborative projects with industries and academics with fullest potential.

PROGRAM EDUCATIONAL OBJECTIVES (PEO'S)

- **PEO1-** Solve real world problems using advanced computing technologies.
- **PEO 2-** Encompass the ability to build innovative software products and become a successful entrepreneur.
- **PEO 3-** Carry out industrial projects and succeed in research careers.

PROGRAM OUTCOMES [PO's]

Engineering students will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems; design systems/components/processes meeting specified needs with safety, societal, and environmental considerations.
4. **Investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Engineering Tool Usage (Modern tool usage):** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. **The engineer and the world:** A merged outcome combining societal, environmental, sustainability concerns; graduates apply reasoning to assess societal, health, safety, legal, cultural, and environmental impacts of engineering solutions.

7. **Ethics and Inclusive Professional Architecture:** Apply ethical principles and commit to professional ethics, responsibilities, inclusive and value-based behavior in engineering practice.

8. **Individual and Collaborative team work:** Function effectively as an individual, and as a member or leader in diverse and multidisciplinary teams, with inclusive and collaborative behavior.

9. **Communication and Global Awareness:** Communicate effectively on complex engineering activities with the engineering community and society at large: writing reports/design docs, effective presentations, clear instructions; with sensitivity to global and inclusive context.

10. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments consider economic and Financial aspects

11. **Life-long learning Adaptability and Tech Awareness:** Recognize need for, and have preparation and ability to engage in independent and lifelong learning, keeping up with evolving technology, global standards, and professional growth.

PROGRAM SPECIFIC OUTPUTS (PSO'S):

PSO1: To apply the knowledge of computer programming to develop the software projects.

PSO2: To design software in futuristic approach to adapt cutting edge technologies.

PSO3: To solve societal problems using computation intelligence.

PO'S & PSO'S ATTAINMENT: *(Put tick Mark for the attained PO's& PSO's)*

PO1	**Engineering Knowledge**		**PO7**	**Ethics and Inclusive Professional Architecture**		**PSO1**	**Apply Computer Programming**	
PO2	**Problem Analysis**		**PO8**	**Individual and Collaborative team work**		**PSO2**	**Design Software**	
PO3	**Design & Development**		**PO9**	**Communication Skills**		**PSO3**	**Solve societal Problems**	
PO4	**Investigations**		PO10	**Project Mgt. & Finance**				
PO5	**Modern Tools**		**PO11**	**Life Long Learning**				
PO6	**Engineer & Society**							

SUSTAINABLE DEVELOPMENT GOALS ATTAINED: (SDG's)



SDG's ATTAINED:

SDG's *(Put tick Mark for the attained SDG's)*																
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**
						✓		✓		✓	✓	✓				

ABSTRACT

This project proposes a non-electrical, self-adjusting solar tracking mechanism that operates using the thermal properties of Phase Change Materials (PCM). The expansion and contraction of PCM during its solid liquid phase change drive a piston mechanism that automatically adjusts the solar panel's tilt angle. The system works entirely without motors, sensors, or electronic controls. It operates 100% without external power, making it a low-cost, sustainable, and maintenance-free solution. The design is especially suitable for a simple and reliable passive mechanism.

Key Words: Phase Change Material (PCM), Solar Tracking, Passive Mechanism, Thermal Expansion, Renewable Energy, Off-Grid System.

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CHAPTER 1
INTRODUCTION

1.1 Overview of Design Thinking:

Design thinking is a systematic and human centered approach used to solve complex problems and develop innovative solutions by focusing on user needs and practical implementation. It is widely applied in engineering, especially in mechanical engineering, where designing efficient and reliable systems is essential. Design thinking involves understanding the problem deeply, analysing user requirements, generating ideas, creating prototypes, and testing solutions. This approach helps engineers move beyond traditional problem-solving methods and encourages creativity and innovation. Instead of focusing only on technical aspects, design thinking emphasizes usability, cost, and real-world application. In the context of this project, design thinking plays a crucial role in identifying the limitations of existing solar tracking systems and developing an alternative solution that is simple, cost-effective, and efficient. By applying design thinking principles, the project focuses on creating a system that addresses real user problems such as high cost, maintenance issues, and dependency on electrical components, thereby ensuring that the final solution is practical and beneficial.

1.2 Importance of Design Thinking in Mechanical Engineering:

Design thinking is highly important in mechanical engineering because it helps engineers design systems that are not only technically sound but also user-friendly and economically feasible. Mechanical engineering involves the design and development of machines, mechanisms, and systems that must operate efficiently under various conditions. By applying design thinking, engineers can better understand user requirements and develop solutions that meet those needs

effectively. It encourages an iterative approach where ideas are continuously improved based on testing and feedback. This leads to the development of more reliable and efficient systems. Design thinking also promotes innovation by allowing engineers to explore new ideas and unconventional solutions. In this project, design thinking is used to develop a passive solar tracking system that eliminates the need for electrical components. This approach ensures that the system is simple, affordable, and suitable for real-world applications, especially in rural and remote areas where technical support and resources are limited.

1.3 Project Objectives:

The main objective of this project is to design and develop a PCM-based passive solar tracking system that improves the efficiency of solar panels without using electrical components. The system aims to automatically adjust the position of the solar panel to follow the movement of the sun, thereby maximizing solar energy absorption. One of the key objectives is to eliminate the use of motors, sensors, and controllers, which are commonly used in conventional tracking systems and contribute to higher cost and complexity. The project also focuses on utilizing Phase Change Material (PCM) to convert thermal energy into mechanical motion, enabling automatic tracking. Another objective is to reduce the overall cost of the system so that it can be easily adopted by small-scale users and rural communities. Additionally, the system is designed to require minimal maintenance and provide reliable performance under different environmental conditions. By achieving these objectives, the project aims to provide a practical and sustainable solution for improving solar energy utilization.

1.4 Scope and Limitations of the Project:

The scope of this project includes the design, fabrication, and analysis of a PCM-based passive solar tracking system that can operate without electrical components. The system is intended to improve the efficiency of solar panels by automatically adjusting their position based on sunlight. The project focuses on developing a simple and cost-effective solution that can be used in rural and off-grid areas where access to electricity and technical support is limited. It also aims to promote the use of renewable energy by providing an eco-friendly alternative to conventional systems. The scope covers material selection, mechanical design, fabrication process, and performance evaluation of the system. However, the project has certain limitations.

The performance of the system depends on environmental factors such as sunlight intensity and temperature. During cloudy conditions or low sunlight, the expansion of PCM may not be sufficient to produce the required movement. Additionally, the system may not provide highly accurate tracking compared to advanced active systems. Despite these limitations, the project offers a practical and efficient solution for improving solar panel performance while maintaining low cost and simplicity.

CHAPTER 2
EMPATHIZE – PROBLEMIDENTIFICATION & USER STUDY

2.1 Background of the Problem:

In recent years, the use of solar energy has increased due to the need for clean and renewable energy sources. However, the efficiency of solar panels depends on the amount of sunlight they receive, and fixed solar panels are not able to capture maximum sunlight throughout the day because the position of the sun changes continuously. To overcome this issue, solar tracking systems are used to adjust the panel position, but conventional tracking systems use electrical components such as motors and sensors, which increase cost and complexity. These systems also require maintenance and consume some amount of power for their operation, making them less suitable for rural and remote areas. Therefore, there is a need for a simple and low-cost system that can improve solar panel efficiency without using electrical components, which leads to the development of a PCM-based passive solar tracking system.

2.2 Identification of Target Users / Stakeholders:

The system is mainly designed for users who require a low-cost and reliable solar energy solution, especially in rural and remote areas. Farmers, small households, and small-scale industries are the primary users who depend on solar energy due to limited electricity supply. These users need a system that is simple to use, affordable, and requires minimal maintenance. In addition to users, stakeholders include engineers, researchers, and organizations working in renewable energy, as well as government bodies that promote sustainable energy solutions. The system is designed to meet the needs of these users by providing an efficient and easy-to-use solar tracking mechanism.

2.3 User Research Methodology:

User research is carried out to understand the real problems faced by users and to design a suitable solution. Different methods such as interviews, surveys, and observations are used to collect information. These methods help in identifying user needs, difficulties, and expectations from solar systems. By analysing the collected data, it becomes easier to understand the key issues and develop a system that addresses these problems effectively.

2.3.1 Interviews:

Interviews are conducted with users like farmers and solar panel users to understand their experiences. Many users reported that fixed panels do not provide sufficient energy output and that advanced tracking systems are costly and difficult to maintain. These interactions provide useful insights into practical problems faced by users.

2.3.2 Surveys:

Surveys are used to collect information from a larger group of users through questionnaires. The results show that most users prefer a low-cost, simple, and reliable system. They also highlight the importance of low maintenance and easy operation.

2.3.3 Observations:

Observations involve studying how solar panels are used in real environments. It helps in identifying issues such as improper alignment, low efficiency, and environmental effects like sunlight variation. This method provides practical understanding of the problem.

2.4 User Needs and Pain Points:

From the study, it is clear that users need a system that is affordable, simple, and efficient. The major problems include high cost of existing systems, complexity in operation, and maintenance issues. Users also face reliability problems due to failure of electrical components. These pain points highlight the need for a better solution.

2.5 User Personas:

User personas represent different types of users. For example, a farmer needs a low-cost system for irrigation, while a small industry owner needs a reliable energy source. These personas help in understanding user requirements clearly and designing a suitable system.

2.6 Empathy Map:

An empathy map helps in understanding user behaviours by analysing what they think, feel, say, and do. Users often feel frustrated due to high cost and low efficiency. They think about improving energy output but are limited by resources. This analysis helps in designing a better solution.

2.7 Key Insights from Empathy Study:

The study shows that users prefer simple, low-cost, and maintenance-free systems. They are more interested in practical solutions that do not require technical knowledge. These insights lead to the development of a PCM-based passive solar tracking system.

CHAPTER 3
DEFINE-PROBLEM IDENTIFICATION

3.1 Analysis of User Research Findings:

The user research findings reveal that most users are dissatisfied with the low efficiency of fixed solar panels because the panel receives maximum sunlight only for a short period of time. Rural users and small-scale industries require a low-cost solar tracking system, but existing motor-based tracking systems are expensive and need continuous electrical power and maintenance. Users also expressed that they prefer a system that is simple, easy to install, and capable of operating automatically without complicated controls. The research further indicates that the major difficulty in existing systems is the dependency on sensors, motors, and electronic circuits, which increase the overall cost and make the system unsuitable for remote areas.

Therefore, there is a need for a passive solar tracking system that can improve panel orientation using simple mechanical and thermal principles. Passive tracking systems are considered more suitable because they do not consume electrical energy and can improve solar energy collection by up to 23% compared to fixed panels.

3.2 Problem Statement:

The major problem identified in this project is that conventional fixed solar panels are unable to continuously face the sun throughout the day. Because of this, the amount of sunlight received by the panel decreases as the sun moves, resulting in reduced energy output and poor efficiency. Although active solar tracking systems are available, they use motors, sensors, and controllers that increase cost, power consumption, and maintenance. These systems are also not suitable for rural

and remote areas where electricity and technical support are limited. Hence, there is a requirement for a low-cost, energy-efficient, and easy-to-maintain passive solar tracking system that can automatically adjust the panel position without using any electrical components. The proposed PCM-based passive solar tracking system is intended to solve this problem by using the thermal expansion property of Phase Change Material to create the movement required for tracking.

3.3 "How Might We (HMW)" Questions:

"How Might We" questions are used to convert the identified problem into opportunities for solution development. In this project, the following HMW questions are framed to guide the ideation process: How might we improve the efficiency of solar panels without using electrical energy? How might we design a low-cost tracking system that can be used in rural areas? How might we use the thermal properties of Phase Change Material to create automatic movement of the solar panel? How might we reduce the dependence on motors, sensors, and controllers in solar tracking systems? How might we make the system simple, reliable, and easy to maintain? These questions help in generating innovative ideas and direct the project toward developing a practical passive tracking solution. HMW questions are considered an important part of the define stage because they transform a problem into a broader opportunity for innovation and creative thinking.

3.4 Design Constraints and Assumptions:

The design of the proposed system is developed by considering certain constraints and assumptions. The main design constraint is that the system should operate without using any electrical components such as motors, sensors, or controllers. The system must be low-cost and easy to manufacture using locally

available materials. The size and weight of the solar panel should be within the capacity of the piston and linkage mechanism. The selected PCM should have a suitable melting temperature so that it can expand under normal sunlight conditions. Improper selection of PCM may reduce the efficiency of the system or cause failure in movement.

Another constraint is that the system performance depends on sunlight intensity and ambient temperature, so the movement may be less during cloudy weather. It is assumed that the system will be installed in an open area where sufficient sunlight is available throughout the day and that the mechanical components are properly aligned to ensure smooth movement. The design also assumes that the PCM container is properly insulated to reduce heat loss and improve performance.

CHAPTER 4
IDEATE – IDEA GENERATION & SELECTION

4.1 Ideation Techniques Used:

The ideation stage is the third step in the design thinking process where various possible solutions are generated based on the problems identified in the previous stages. After understanding user needs and defining the core problem, this stage focuses on creating multiple ideas without any limitations. The main purpose of ideation is to explore a wide range of solutions and encourage creative thinking. It is considered an important phase because it helps in identifying innovative and practical solutions to the problem.

In this project, the ideation stage is used to generate different concepts for improving solar panel efficiency without using electrical components, leading to the development of a PCM-based passive solar tracking system. The goal of this stage is not to find one solution immediately but to produce many ideas and then select the most suitable one.

4.2 Generation of Multiple Concepts:

Brainstorming is the main activity in the ideation stage where multiple ideas are generated freely. In this process, all possible solutions are considered without judging whether they are practical or not. The focus is on generating a large number of ideas, as this increases the chances of finding innovative solutions. Some of the ideas generated for this project include using mechanical systems, hydraulic systems, thermal expansion methods, and passive tracking mechanisms. Among these, the idea of using Phase Change Material was found to be more suitable because it can convert thermal energy into mechanical motion.

Brainstorming also helps in combining and improving ideas, which leads to better solutions.

4.3 Concept screening and Evaluation Criteria:

After generating multiple ideas, the next step is to develop concepts based on the most promising solutions. Concept development involves analysing different ideas and selecting the ones that best meet the requirements of simplicity, cost-effectiveness, and efficiency. In this project, different concepts such as motor-based tracking, spring-based mechanisms, and fluid expansion systems were evaluated.

Among these, the PCM-based system was selected because it does not require electrical components and provides automatic operation. The concept is developed by designing a mechanism where the expansion of PCM is used to move a piston, which in turn adjusts the solar panel position. This concept ensures efficient solar tracking with minimal complexity.

4.4 Selection of Final concept:

The selection of the best idea is based on various factors such as feasibility, cost, reliability, and ease of implementation. Each concept generated during brainstorming is analysed and compared. The PCM-based passive solar tracking system is selected because it meets all the required criteria.

It is simple in design, does not require electrical components, and can operate automatically using solar heat. It is also cost-effective and suitable for rural applications. The selection process ensures that the final solution is practical and efficient.

4.5 Justification of Selected Concept:

The selected PCM-based system is justified based on its advantages over other methods. Unlike motor-based systems, it does not consume electrical energy and requires less maintenance. Compared to complex mechanical systems, it is simpler and easier to fabricate. The use of PCM allows the system to utilize natural heat energy, making it environmentally friendly. The system is also reliable because it has fewer components, reducing the chances of failure. These factors make the selected concept more suitable for real-world applications.

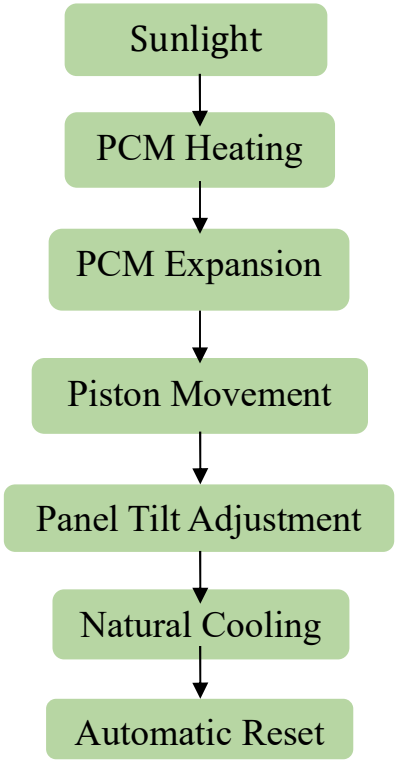
CHAPTER 5
SELECTION OF MATERILS

5.1 Working Principle of the Proposed Solution:

The proposed PCM-based passive solar tracking system works on the principle of thermal expansion of Phase Change Material. When sunlight falls on the PCM container, the material absorbs heat and changes its phase from solid to liquid, resulting in expansion. This expansion creates pressure inside the container, which pushes the piston connected to it. The piston movement is then transferred to a linkage mechanism that adjusts the position of the solar panel. As the sun moves, continuous heating causes gradual movement of the panel towards the sunlight. During evening, when temperature decreases, the PCM contracts and returns to its original state, causing the panel to reset. This cycle continues automatically without using any electrical components.

5.2 Functional Block Diagram:

The functional block diagram of the system represents the overall working flow of the proposed model. The main input to the system is solar energy, which is absorbed by the PCM. The PCM acts as the energy conversion unit, converting thermal energy into mechanical motion. This motion is transmitted through a piston-cylinder mechanism, which is connected to a linkage system. The linkage system controls the movement of the solar panel. The output of the system is the continuous adjustment of the solar panel position to maximize sunlight exposure. The block diagram helps in understanding the flow of energy and operation of each component in the system.



5.3 Mechanical Design Description:

The mechanical design of the system consists of a supporting frame, PCM container, piston-cylinder arrangement, linkage mechanism, and solar panel mounting structure. The frame is designed to provide stability and support all components. The PCM container is fixed securely and connected to the piston. The piston is designed to move smoothly inside the cylinder with proper sealing to prevent leakage. The linkage mechanism is connected between the piston and the solar panel, converting linear motion into angular motion. The solar panel is mounted in such a way that it can tilt easily according to the movement. The overall design is simple, compact, and easy to fabricate.

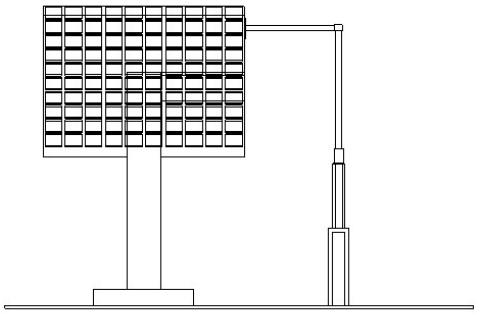
5.4 Material Selection:

Material selection is an important factor in the design of the system to ensure durability, efficiency, and cost-effectiveness. Mild steel is selected for the frame

due to its strength and availability. The PCM is chosen based on its ability to absorb and release heat effectively within a suitable temperature range. The piston and cylinder are made using materials that provide smooth movement and resistance to wear. Fasteners like nuts and bolts are used for assembly. The selected materials are economical and suitable for the working conditions of the system.

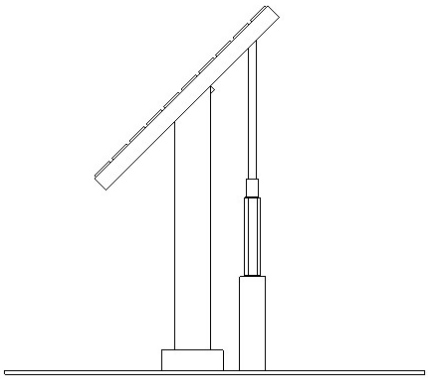
5.5 CAD Models:

CAD models are created to design and visualize the system before fabrication. The CAD design includes all components such as the frame, PCM container, piston mechanism, and solar panel mounting. These models help in understanding the dimensions, alignment, and assembly of the system. CAD modelling also helps in identifying design errors and making necessary modifications before actual fabrication. It ensures accuracy and improves the overall design quality.



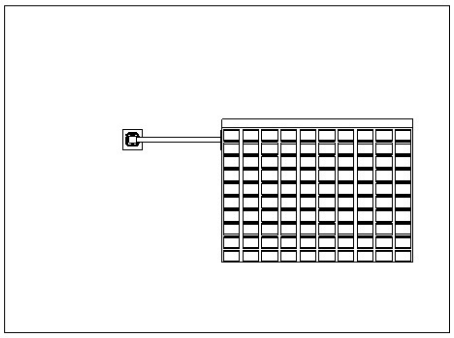
Front view of solar tracking system

Fig- 5.5.1



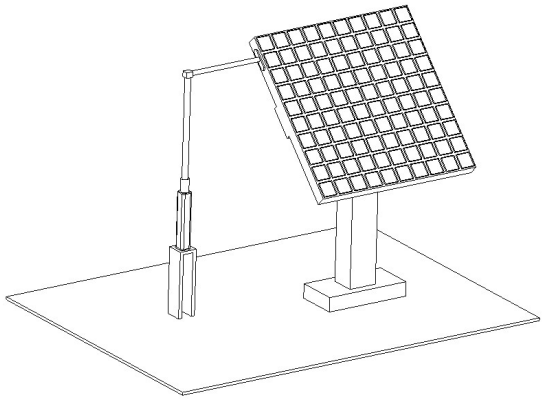
Side view of solar tracking system

Fig- 5.5.2



Top view of solar tracking system

Fig- 5.5.3

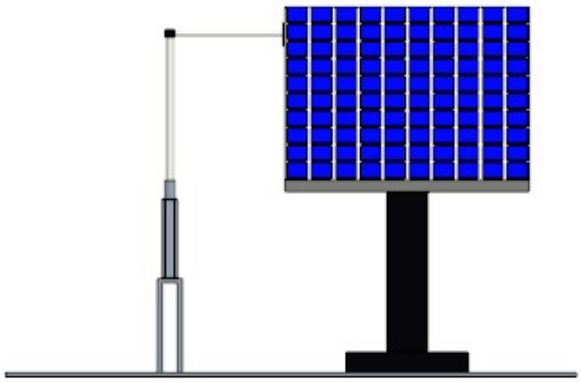


Isometric view of solar tracking system

Fig- 5.5.4

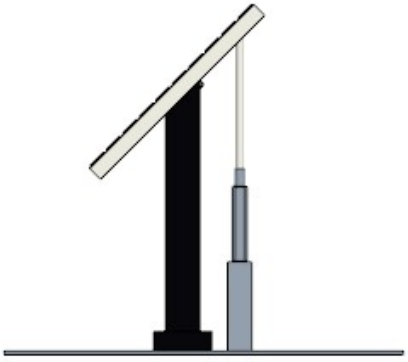
5.6 Virtual Modelling:

Virtual modelling is used to simulate the working of the system using software tools. It helps in analysing the movement of components and understanding how the system behaves under different conditions. Through simulation, the expansion of PCM and movement of the piston can be studied. Virtual modelling reduces errors and improves design efficiency by allowing modifications before physical implementation. It also saves time and cost in the development process.



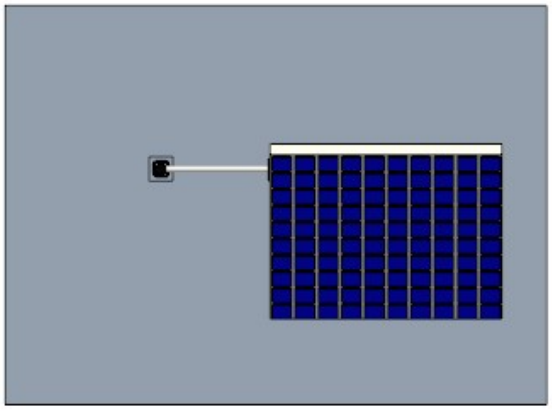
Front view of solar tracking system

Fig- 5.6.1



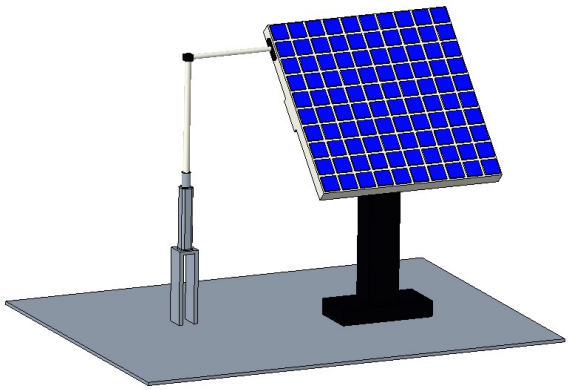
Side view of solar tracking system

Fig- 5.6.2



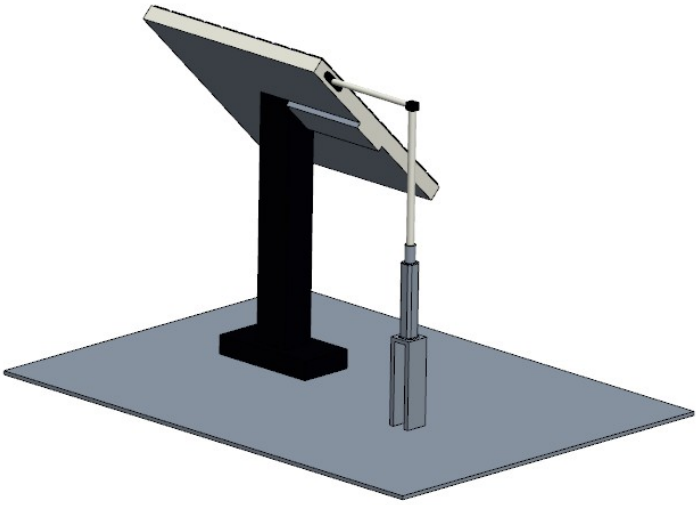
Top view of solar tracking system

Fig- 5.6.3



Isometric view of solar tracking system-1

Fig- 5.6.4



Isometric view of solar tracking system-2

Fig- 5.6.5

5.7 Components and Tools Used:

Table- 5.7.1

S. No	Component / Tool Name	Purpose
1	Solar Panel	Converts sunlight into electrical energy
2	PCM (Phase Change Material)	Absorbs heat and expands to create motion
3	PCM Container	Holds the PCM material
4	Piston and Cylinder	Converts expansion into linear motion
5	Linkage Mechanism	Transfers motion to solar panel
6	Supporting Frame (Mild Steel)	Provides structural support
7	Nuts and Bolts	Used for assembly of components
8	Welding Machine	Used for joining metal parts
9	Cutting Tools	Used for shaping materials
10	Measuring Tools	Ensures accurate dimensions

CHAPTER 6
TEST – TESTING, RESULTS AND VALIDATION

6.1 Testing Methodology:

The testing methodology of the PCM - based passive solar tracking system involves evaluating the performance of the prototype under real environmental conditions. The system is placed in an open area where it is exposed to natural sunlight throughout the day. The behaviour of the PCM, movement of the piston, and tilt of the solar panel are observed at different time intervals. The testing is conducted during various periods such as morning, noon, and evening to analyse how the system responds to changes in sunlight intensity and temperature. All observations are recorded systematically to understand the working efficiency of the system. This method helps in analyzing the real-time performance and identifying any issues in the system.

6.2 Performance Evaluation:

The performance of the system is evaluated based on its ability to track the sun and improve the efficiency of the solar panel. The movement of the panel is checked to ensure that it aligns properly with the direction of sunlight. The smoothness of piston movement and effectiveness of the linkage mechanism are also considered. The system is compared with a fixed solar panel to observe the difference in alignment and energy capture. Stability, reliability, and consistency of operation are also evaluated to ensure that the system performs efficiently under different conditions.

6.3 User Testing and Feedback:

User testing is carried out by allowing users to observe and interact with the system. Feedback is collected from users such as students, technicians, and individuals familiar with solar systems. Most users find the system simple, easy to understand, and cost-effective. They appreciate the absence of electrical components and the automatic operation of the system. Some users suggest improvements such as better insulation and smoother mechanical movement. This feedback is useful in identifying practical issues and improving the design of the system.

6.4 Results and Observations:

The results show that the PCM - based passive solar tracking system operates effectively under normal sunlight conditions. The PCM absorbs heat and expands as expected, generating sufficient force to move the piston and adjust the solar panel position. The system shows gradual and continuous movement, allowing the panel to follow the sun throughout the day. It is observed that the system performs better than a fixed panel in terms of alignment with sunlight. However, during cloudy conditions or low sunlight, the movement is reduced due to decreased heat absorption. Overall, the observations indicate that the system is efficient, reliable, and suitable for practical applications.

CHAPTER 7
FEASIBILITY & SUSTAINABILITY ANALYSISGIVEN DATA

7.1 Cost Analysis:

Cost analysis is an important factor in evaluating the feasibility of the proposed PCM - based passive solar tracking system. It includes the estimation of initial investment cost, fabrication cost, and maintenance cost. The initial cost mainly involves materials such as mild steel, PCM, piston-cylinder components, and structural elements. Fabrication cost includes cutting, welding, and assembly processes, which are relatively low due to the simple design of the system. Maintenance cost is also minimal because the system does not use electrical components like motors or sensors, which are prone to failure. Compared to conventional tracking systems, the proposed system is more economical, as active systems usually have higher installation and operational costs due to electrical and control components.

Studies show that although tracking systems may increase initial cost, they improve energy output and reduce long-term energy cost, making them economically beneficial. Thus, the proposed system provides a cost-effective solution with low investment and maintenance requirements.

7.2 Manufacturability:

Manufacturability refers to the ease with which the system can be fabricated using available resources and techniques. The proposed system is designed with simple mechanical components that can be easily manufactured using standard

workshop tools. Materials such as mild steel and basic mechanical parts are readily available in the market, which simplifies the manufacturing process. The design avoids complex machining and does not require specialized equipment, making it suitable for small-scale production. The assembly process is straightforward, involving basic operations such as cutting, welding, and fastening. This simplicity ensures that the system can be produced at low cost and with minimal technical expertise. The manufacturability of the system makes it practical for local fabrication and widespread implementation.

7.3 Safety Considerations:

Safety is an essential aspect in the design and implementation of any mechanical system. In this project, safety considerations include structural stability, proper handling of materials, and safe operation of mechanical components. The supporting frame is designed to withstand environmental conditions such as wind and load.

The piston and linkage mechanisms are properly aligned to avoid mechanical failure or accidents. Since the system does not use electrical components, the risk of electrical hazards is eliminated. During fabrication, safety measures such as proper use of tools, protective equipment, and safe welding practices must be followed. Ensuring proper installation and regular inspection of components helps in maintaining safe operation of the system over time.

7.4 Environmental and Sustainability Aspects:

The proposed PCM-based passive solar tracking system is environmentally friendly and supports sustainable energy usage. Solar energy is a clean and renewable source that reduces dependence on fossil fuels and minimizes pollution. The system improves energy efficiency by maximizing solar energy capture,

thereby reducing the number of panels required. Studies indicate that solar tracking systems can significantly enhance energy output and contribute to reduced carbon emissions. Additionally, the use of simple mechanical components reduces electronic waste and energy consumption. However, manufacturing processes involve materials like steel and aluminium, which have environmental impacts such as CO_2 emissions during production.

Despite this, the long-term benefits of renewable energy and reduced emissions make the system sustainable. Overall, the system promotes eco-friendly energy generation and supports sustainable development.

CHAPTER 8
CONCLUSION & FUTURE SCOPE

8.1 Conclusion:

The project titled “PCM-Based Passive Solar Tracking System” has been successfully designed, developed, and validated by applying the principles of design thinking. The system effectively solves the problem of reduced efficiency in fixed solar panels by automatically adjusting their position according to sunlight without using any electrical components. The use of Phase Change Material (PCM) enables the conversion of thermal energy into mechanical motion, which drives the movement of the solar panel. This makes the system simple, cost-effective, and energy-efficient. Compared to conventional tracking systems, the proposed system eliminates the need for motors, sensors, and controllers, thereby reducing cost, maintenance, and power consumption.

Passive solar tracking systems are known to improve energy collection efficiency significantly and operate without external power sources, making them sustainable and reliable. The prototype testing and implementation results show that the system is capable of following the sun’s movement and improving alignment, which leads to better energy utilization. Although the system has some limitations under low sunlight conditions, the overall performance is satisfactory.

The project demonstrates that simple mechanical and thermal principles can be effectively used to develop innovative solutions for renewable energy applications. Thus, the proposed system is a practical and eco-friendly solution that supports sustainable energy development.

8.2 Learning Outcomes from Design Thinking Process:

The design thinking process followed in this project provides valuable learning outcomes in both technical and practical aspects. It helps in understanding how to approach real-world problems in a structured and systematic manner by focusing on user needs and problem identification. Through the empathize stage, the importance of understanding user requirements and challenges is learned. The define stage helps in clearly identifying the core problem and its causes. The ideate stage encourages creative thinking and generation of multiple solutions, which improves innovation skills.

The prototype stage provides hands-on experience in designing and developing a working model, while the testing stage helps in evaluating performance and identifying improvements. Design thinking is an iterative and solution-focused approach that enhances problem-solving ability, creativity, and decision-making skills. It also promotes teamwork, communication, and practical knowledge, which are essential for engineering applications. Through this project, knowledge about solar energy systems, passive mechanisms, and material behaviour is gained.

The process also highlights the importance of continuous improvement through feedback and testing. Overall, the project provides a complete learning experience in applying engineering concepts to develop real-world solutions.

8.3 Future Scope and Improvements:

The proposed PCM-based passive solar tracking system has significant potential for future development and improvement. One of the major areas of

improvement is the selection of advanced Phase Change Materials with higher thermal efficiency and faster response time, which can enhance system performance even under low sunlight conditions.

Proper insulation techniques can be implemented to reduce heat loss and maintain consistent temperature within the PCM container. The mechanical design can be further optimized to reduce friction and improve smooth movement of the piston and linkage mechanism. The system can also be scaled up to support larger solar panels and used in industrial and commercial applications. Integration of hybrid systems combining passive and active tracking methods can improve accuracy while maintaining low energy consumption.

Research shows that solar tracking systems can increase energy output by 12–20% compared to fixed systems, highlighting the importance of further development in this field. In addition, future work can focus on improving durability, adapting the system for different climatic conditions, and using eco-friendly materials to enhance sustainability. Passive systems are naturally energy-efficient as they operate without external power and rely on natural heat transfer mechanisms. With continuous research and development, the system can become more efficient, reliable, and widely applicable in real-world renewable energy solutions.

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